



# TITLE OF LECTURE: PRINCIPLE OF RADIATION THERAPY

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## OBJECTIVES:

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- Describe medical use of radiation based on the fact that it can destroy cells (instrument sterilization, treatment of cancer).
- To familiarize the student with the basic principles of equipment used for external beam radiotherapy.

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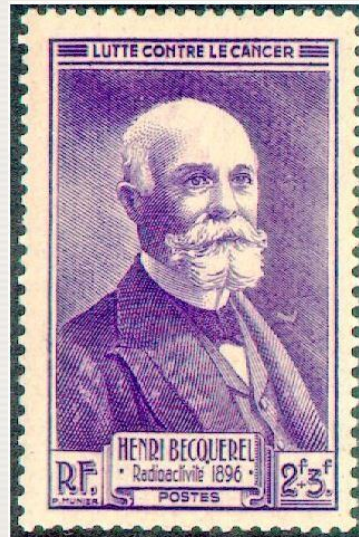
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1. Introduction
2. uses of ionising radiation in medicine
3. X-ray & Gamma ray beams and x-ray units
4. Linacs
5. Radiotherapy with protons, neutrons, and heavy ions
6. Shielding considerations
7. Cobalt-60 teletherapy units versus linacs
8. Radiotherapy Treatment
  - 8.1.Planning
  - 8.2.Simulators and computed tomography simulators
  - 8.3. Different planning for treatment radiotherapy.
9. Radiotherapy protection.

# I. INTRODUCTION

□ The study and use of ionizing radiation in medicine started with three important discoveries:

- X rays by Wilhelm Roentgen in .1895
- Natural radioactivity by Henri Becquerel in .1896
- Radium-226 by Pierre and Marie Curie in .1898





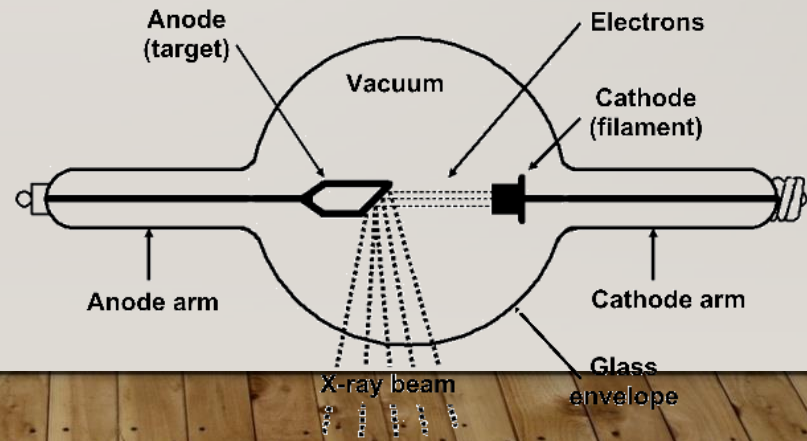
# TERMINOLOGY

|                      |                              |                                                                                                                                                                 |
|----------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Radiation Oncology   | Discipline of human medicine | Concerning the causes, prevention, and treatment of cancer and other diseases involving special expertise in the therapeutic applications of ionizing radiation |
| Radiotherapy         | Clinical modality            | Dealing with the use of ionizing radiations the treatment of patients with malignant tumor (and occasionally benign diseases)                                   |
| Radiation Oncologist | Person                       | Use radiation therapy to try to cure cancer, to control cancer growth or to relieve symptoms, such as pain.                                                     |

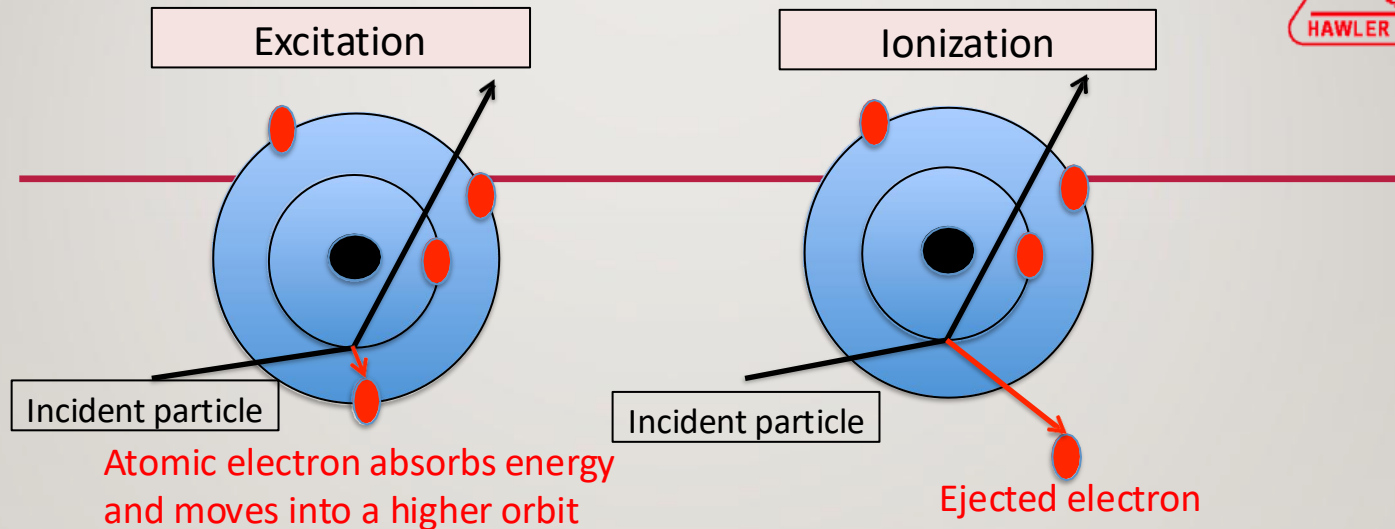
# INTRODUCTION.

❑ Coolidge in 1913 designed a “hot cathode” x ray tube and his design is still in use today.

- The main characteristics of the Coolidge tube are its high vacuum and its use of heated filament (cathode).
- The heated filament emits electrons through thermionic emission.
- X rays are produced in the target (anode) through radiation losses of electrons producing characteristic and bremsstrahlung photons.
- The maximum photon energy produced in the target equals the kinetic energy of electrons striking the target.



# INTRODUCTION.



- Excitation = the incident particle is not sufficient to eject an electron from the atom but is used to raise the electrons to higher- energy levels
- Ionization = the electron is removed, leaving the atom positively charged, resulting in an ion pair

1. Khan F. The Physics of Radiation Therapy

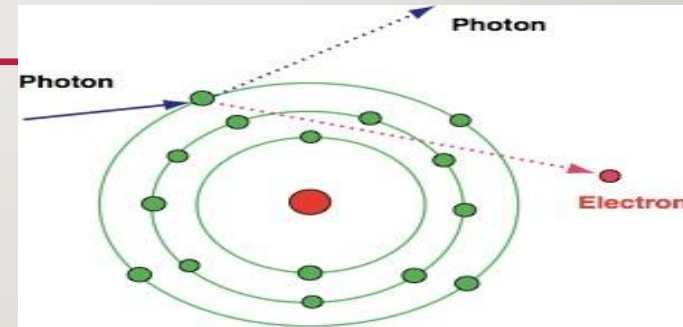
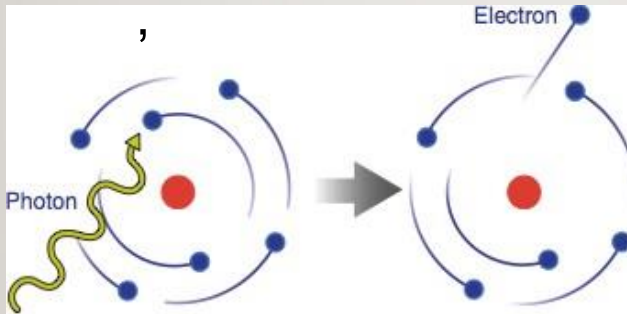
## EXCITATION VS. IONIZATION

# INTRODUCTION.

## INTERACTION RADIATION WITH MATTER

### COMPTON EFFECT

### PHOTOELECTRIC EFFECT



- It is dominant at energies of less than 35 kV, and in atoms with high atomic numbers (Z).
- Since the atomic number of bone is higher than that of soft tissue, bone absorbs more radiation than soft tissue.
- Diagnostic radiology.
- Radiation Oncology
- It is the dominant effect across a wide spectrum of energies, such as 35 kV–50 MV. It has no dependency on the atomic number (Z) of the absorbent material, but it does depend on the electron density of the material.
- The absorption of incoming radiation is the same for bone and soft tissues.

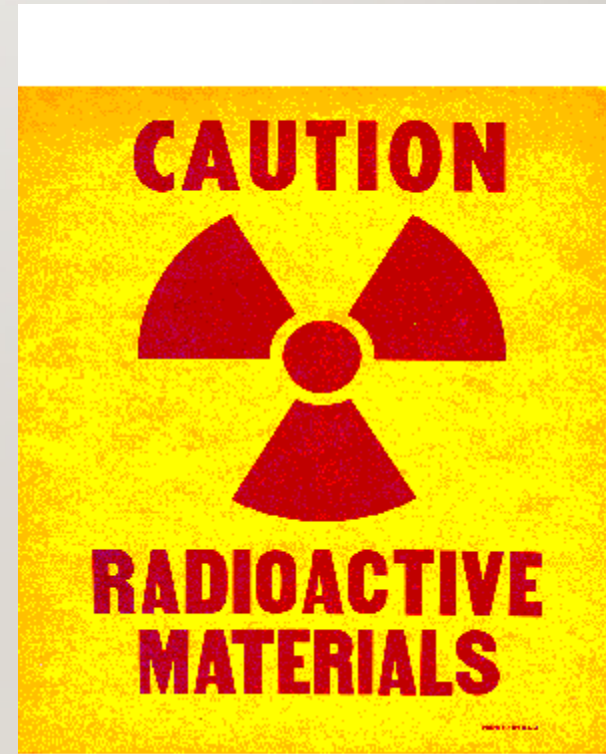
1. Basic Radiation Oncology.



2.

## THERE ARE 3 MAIN USES OF IONISING RADIATION IN MEDICINE:

- Diagnosis
- Treatment
- Sterilisation

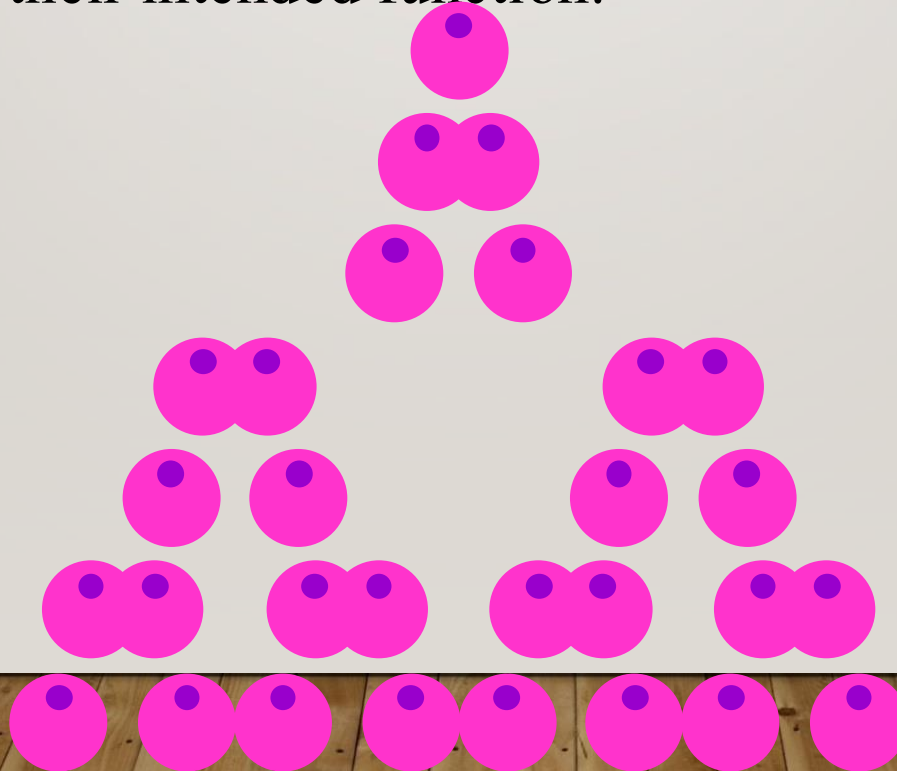


2.

## WHAT IS CANCER?

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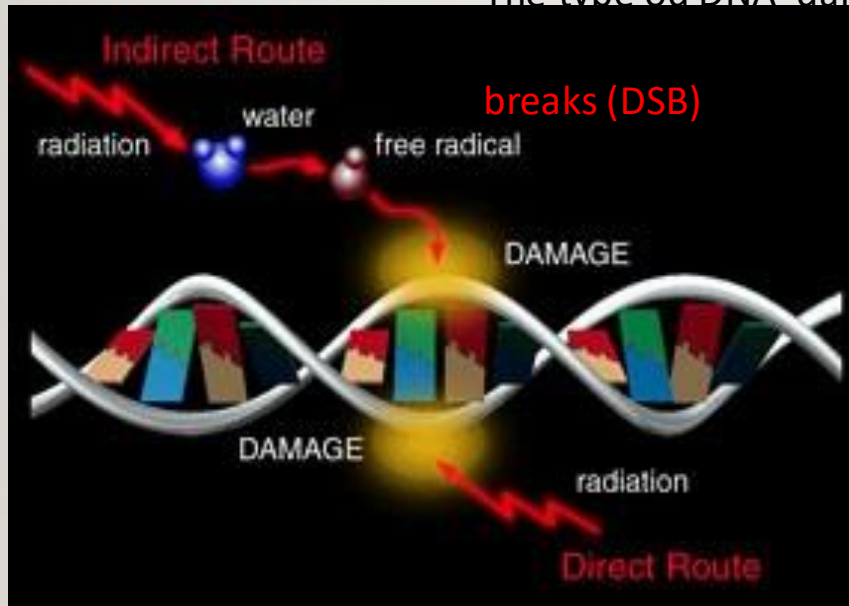
Cancers are growths of cells (cancerous tumours) which are out of control. As a result of this, they do not perform their intended function.



## 2. RADIATION INDUCED DAMAGED

- DNA is primary target
- Double Strand breaks –Primary requisite (irreversible and irreparable damage )
- Reproductive Cell Death →Apoptosis

The type of DNA damage:



- Double strand
- Single strand breaks (SSB)
- Base damage
- Cross links damage

# BIOLOGICAL EFFECTS OF IONIZING RADIATION

At a cellular/tissue level, effects are categorized •  
as **stochastic** or **deterministic (tissue reactions)**

LOW DOSES

Cell mutations



Germ cell



Somatic cell

Hereditary  
effects

Cancer,  
leukaemia

**STOCHASTIC EFFECTS  
(PROBABILISTIC)**

HIGH DOSES

Cell death



Loss of function  
(organ/ system/ organism)

**DETERMINISTIC EFFECTS  
(TISSUE REACTIONS)**



## 2. TREATMENT OF CANCER

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Cancerous tumours can be treated using the following main methods:

- Chemotherapy (drugs).
- Radiation therapy .
- Surgery.

2.

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The choice of treatment depends on a number of factors including:

The size of the tumour. •

The position of the tumour. •

## 2. THE AIMS OF RADIATION THERAPY



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1. The aim of radiation therapy is to cause damage to the cancerous cells whilst minimising the risk to surrounding healthy tissue.
  2. The damage inflicted by radiation therapy causes the cancerous cells to stop reproducing and thus the tumour shrinks.
  3. Unfortunately, healthy cells can also be damaged by the radiation.

2.

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WHY DOES THE AMOUNT OF RADIATION GIVEN TO THE PATIENT HAVE TO BE ACCURATELY CALCULATED?

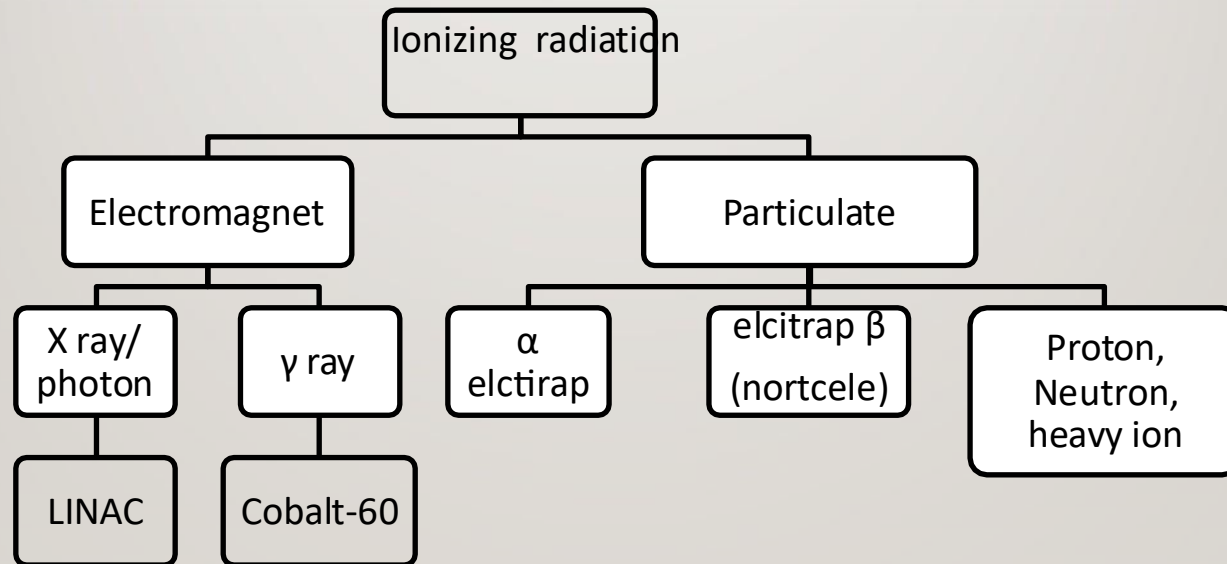
The amount of radiation given to the patient has to be accurately calculated so that the damage is limited to the cancerous cells only.

2.

# DOSIMETRY AND RADIATION PHYSICS



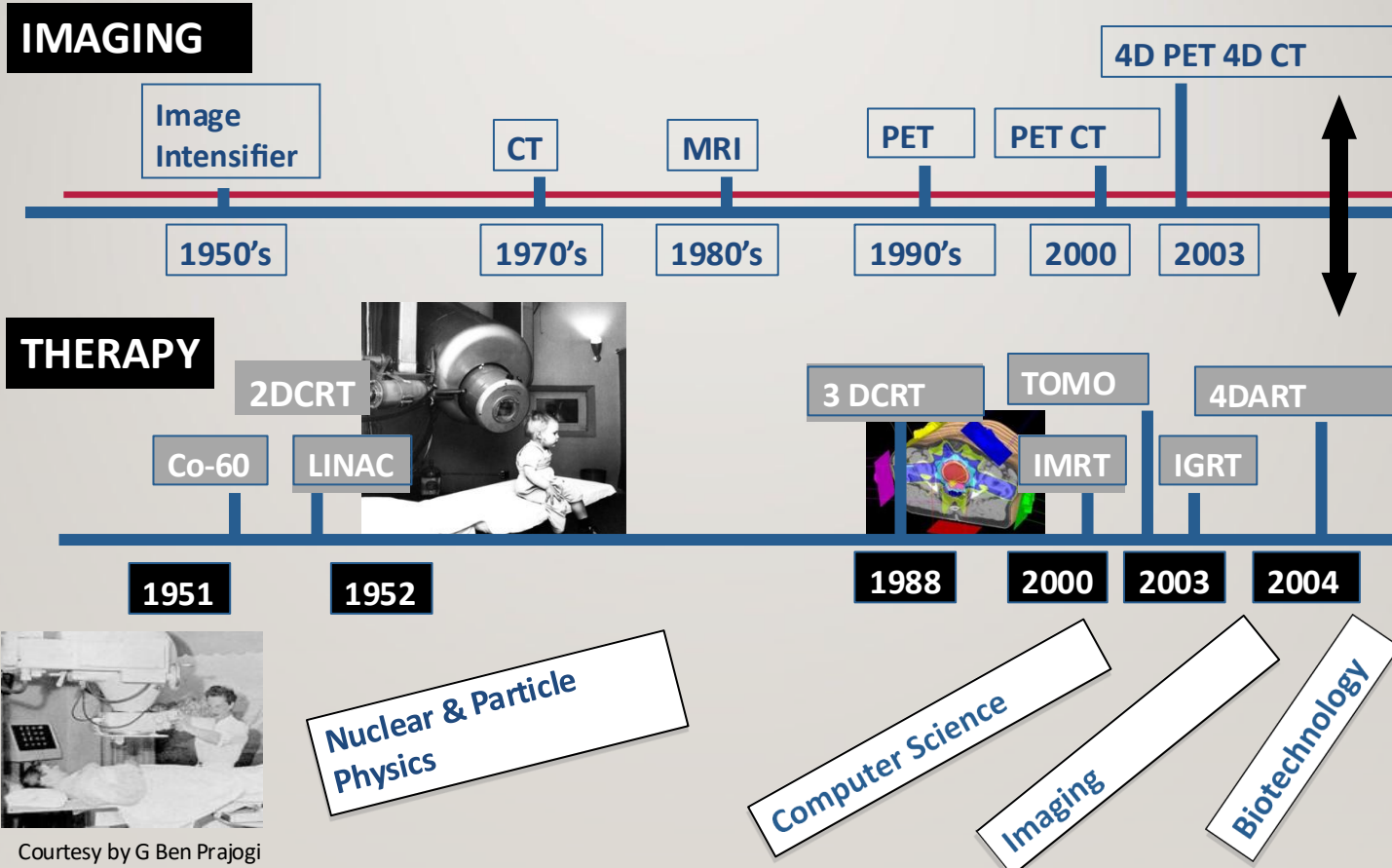
- Dosimetry is the measurement of the absorbed dose delivered by ionizing radiation.



1. Basic Radiation Oncology.



## 2. NEW DEVELOPMENTS OF RADIOTHERAPY



Courtesy by G Ben Prajogi

### 3. X-RAY BEAMS AND X-RAY UNITS

Standard machines used for modern radiotherapy:

- ❑ **X-ray machine:**
  - Superficial x-ray machine: 50 - 80 kVp
  - Orthovoltage x-ray machine: 80 - 350 kVp
- ❑ **Cobalt-60 teletherapy machine**
- ❑ **Linear accelerator (linac):**
  - Megavoltage x rays: 6 - 25 MV
  - Electrons: 6 - 30 MeV

### 3. X-RAY BEAMS AND X-RAY UNITS

Specialized machines used for modern radiotherapy:

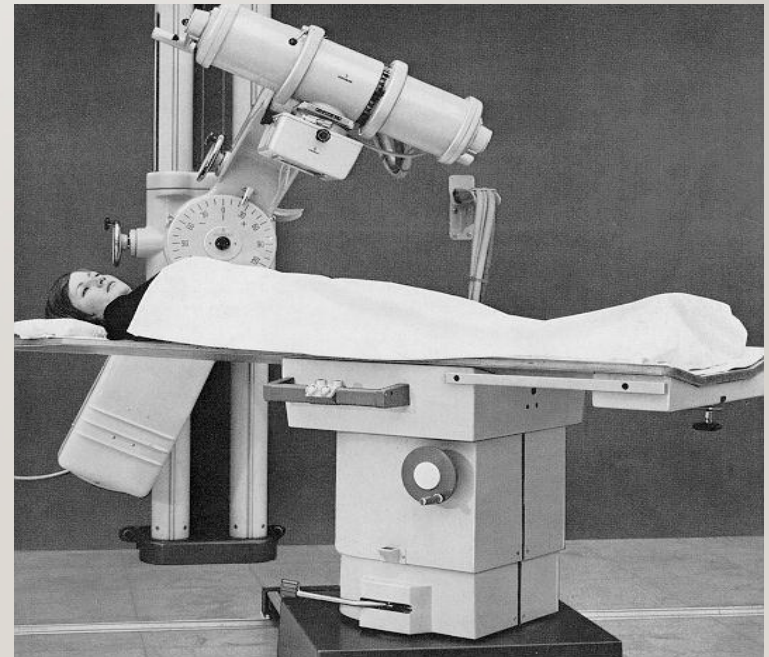
- ☐ Microtron: megavoltage x rays and electrons
- ☐ Betatron: megavoltage x rays and electrons
- ☐ Neutron machines
  - Neutron generator: (d,t) machine producing 14 MeV neutrons
  - Cyclotron accelerating protons
- ☐ Proton machines
  - Cyclotron
  - Synchrotron

### 3. X-RAY BEAMS AND X-RAY UNITS

#### X-ray machines for radiotherapy

□ Superficial and orthovoltage beams used in radiotherapy are produced by **x-ray machines**. The main components of a radiotherapy x-ray machine are:

- X-ray tube
- Ceiling or floor mount for the x-ray tube
- Target cooling system
- Control console





### 3. GAMMA RAY BEAMS AND GAMMA RAY UNITS

#### Basic properties of gamma rays

□ The important characteristics of radionuclides useful for external beam radiotherapy are:

- High gamma ray energy (of the order of 1 MeV).
- High specific activity (of the order of 100 Ci/g).
- Relatively long half life (of the order of several years).
- Large specific air kerma rate constant.

□ Of over 3000 radionuclides known only 3 meet the required characteristics and essentially only cobalt-60 is currently used for external beam radiotherapy.



### 3. GAMMA RAY BEAMS AND GAMMA RAY UNITS

#### Teletherapy machines

Cobalt-60 teletherapy machine, Theratron-780, AECL (now MDS Nordion), Ottawa, Canada



## 4. Linacs

### Linac generations

□ During the past 40 years medical linacs have gone through five distinct generations, each one increasingly more sophisticated:

- (1) Low energy x rays ( 6-4 MV)
- (2) Medium energy x rays (15-10 MV) and electrons
- (3) High energy x rays (25-18 MV) and electrons
- (4) Computer controlled dual energy linac with electrons
- (5) Computer controlled dual energy linac with electrons combined with intensity modulation

## Safety of linac installations

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❑ Safety of operation for the patient, operator, and the general public is of great concern because of the complexity of modern linacs.

❑ Three areas of safety are of interest

- Mechanical
- Electrical
- Radiation

❑ Many national and international bodies are involved with issues related to linac safety.

## Linac treatment head

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☐ **Clinical x-ray beams** are produced with:

- Appropriate x-ray target.
- Appropriate flattening filter.

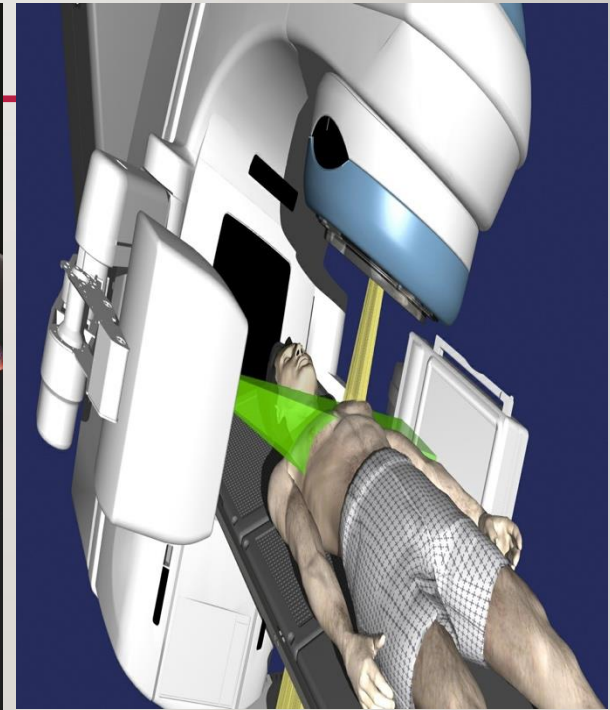
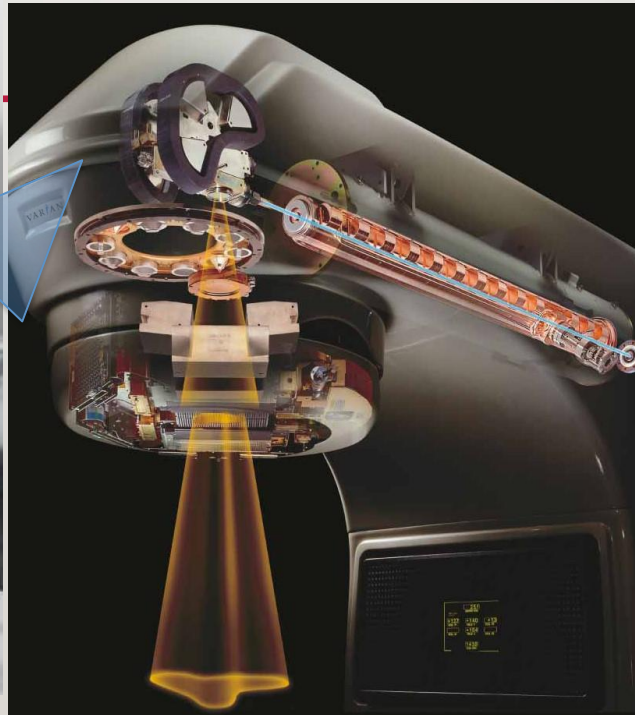
☐ **Clinical electron beams** are produced by:

- Either scattering the pencil electron beam with an appropriate scattering foil.
- Or deflecting and scanning the pencil beam magnetically to cover the field size required for electron treatment.

☐ The flattening filters and scattering foils are mounted on a rotating carousel or sliding drawer.



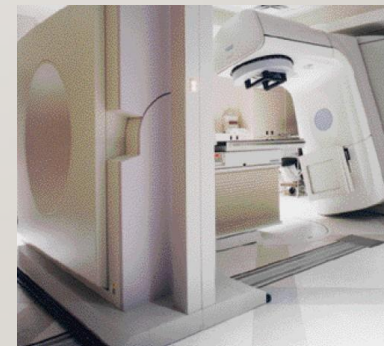
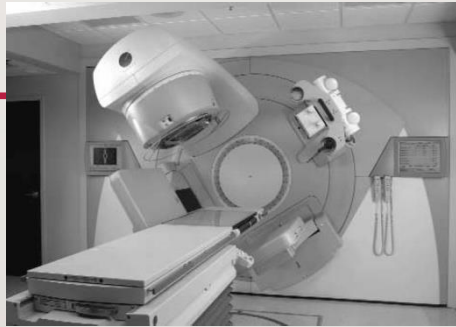
# EXTERNAL BEAM RADIOTHERAPY





# LINEAR ACCELERATORS WITH FEATURES ESPECIALLY SUITABLE FOR SBRT

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\* Slide courtesy of Brian Kavanagh // University of Colorado

## Linac generations

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□ Typical modern dual energy linac, incorporating imaging system and electronic portal imaging device (EPID),  
Elekta, Stockholm



IAEA

# 6. SHIELDING CONSIDERATIONS



- ❑ External beam radiotherapy is carried out mainly with three types of equipment that produces either x rays or electrons for clinical use:
  - X-ray machines (superficial or orthovoltage).
  - Cobalt-60 teletherapy machines.
  - Linacs.
  
- ❑ All radiotherapy equipment must be housed in specially shielded treatment rooms in order to protect personnel and general public in areas adjacent to the treatment rooms.

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❑ Superficial and orthovoltage x-ray therapy rooms are shielded:

- Either with ordinary concrete (density :2.35 ) g/cm Or
- lead (density :11.36 g/cm).

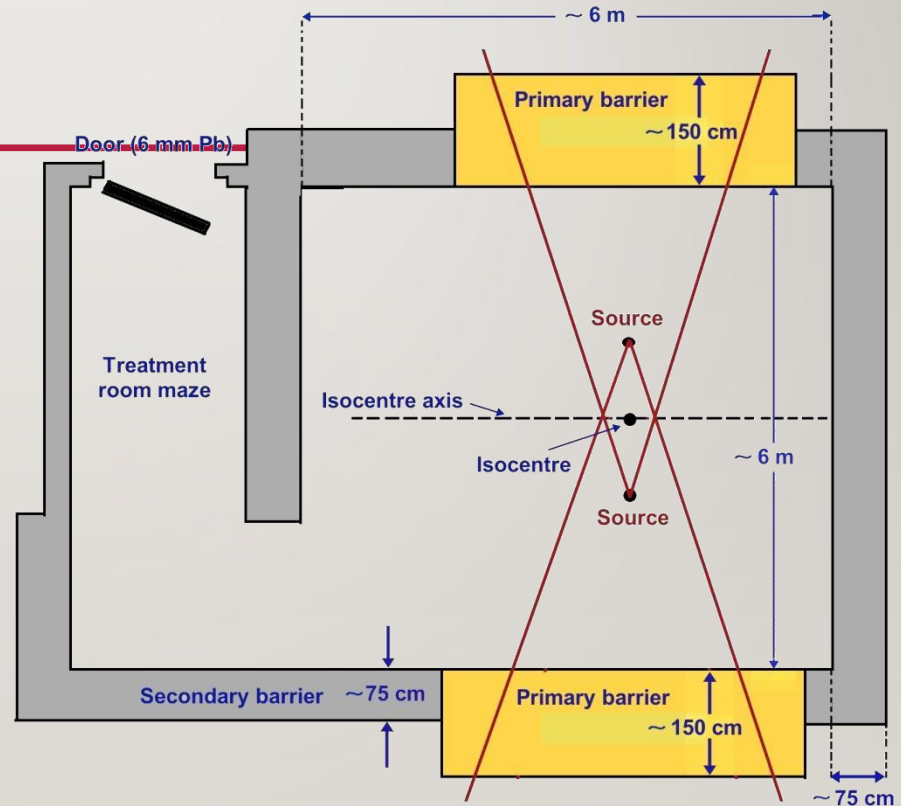
❑ In this energy range the photoelectric effect is the predominant mode of photon interaction with matter, making the use of lead very efficient for shielding purposes.





## Typical cobalt-60 teletherapy installation: isocentric machine

- Primary barriers shield against the primary cobalt-60 beam.
- Secondary barriers shield against leakage radiation and radiation scattered from the patient.





# COBALT-60 TELETHERAPY UNITS VERSUS LINACS

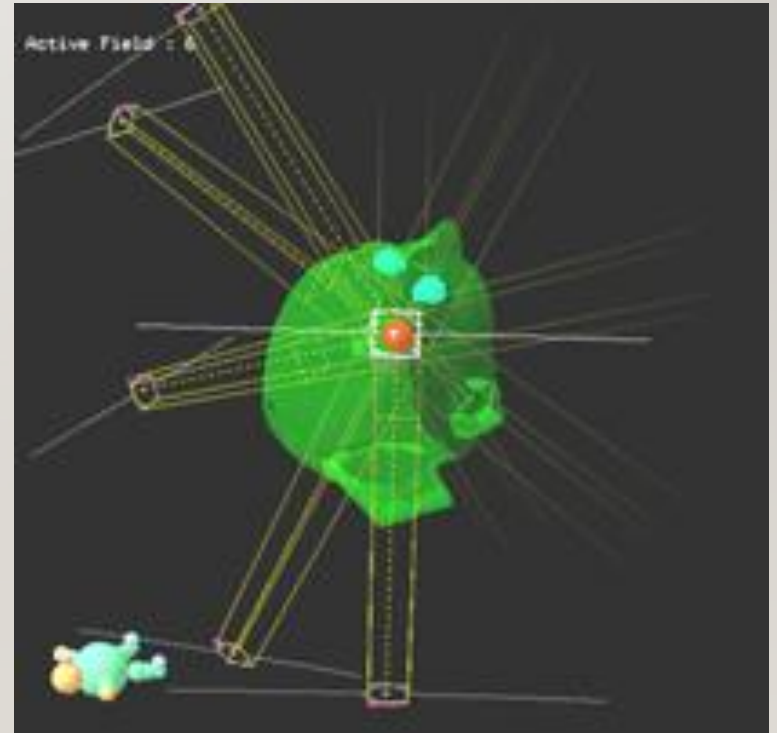
**Modern cobalt-60 teletherapy machine, MDS Nordion, Ottawa, Canada**



## 8. RADIOTHERAPY PLANNING

Every treatment using radiotherapy has to be planned. The planning process consists of three phases:

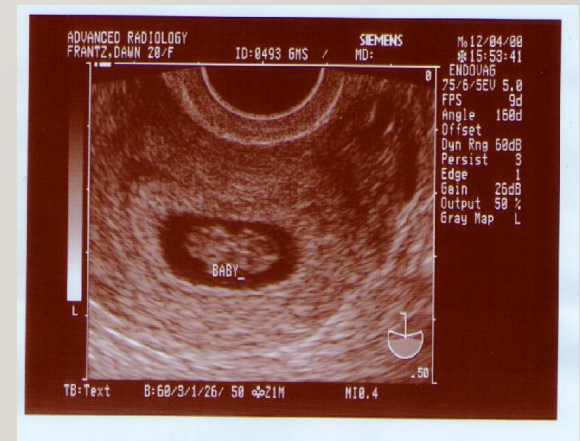
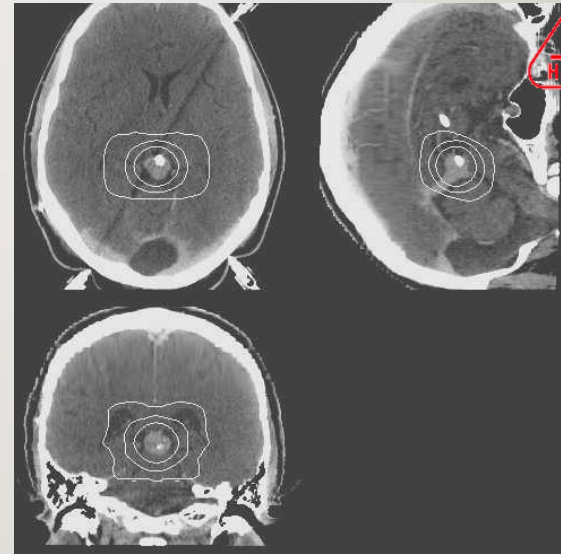
1. Planning
2. Simulation
3. Treatment



## 8.1. PLANNING

The cancerous tumour has to be located so that its size and position can be analysed. This information can be obtained from:

- X-rays
- CT scans
- MRI scans
- Ultrasound images



## 8.2. SIMULATORS AND CT SIMULATORS

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□ Simulators and CT simulators cover several important steps in the radiotherapeutic process related to:

- Determination of target location within the patient.
- Determination of the target shape and volume.
- Determination of the location of critical structures adjacent to treatment volume.
- Planning of dose delivery procedure (treatment planning).
- Accuracy of dose delivery to the target.



## Radiotherapy simulator

□ Radiotherapy simulator consists of a diagnostic x-ray tube mounted on a rotating gantry

to simulate geometries of isocentric teletherapy machines and isocentric linacs.





## Radiotherapy simulator

❑ Modern simulator covers the following processes:

- Tumour and adjacent normal tissue localization.
- Treatment simulation.
- Treatment plan verification.
- Monitoring of treatment.

❑ The design specifications and quality assurance processes for a simulator cover four distinct components:

- Mechanical motions.
- Electrical.
- X-ray tube and generator.
- Image detection.

# SIMULATORS AND CT SIMULATORS

## CT simulator

- ❑ Oncology CT simulator (Philips)
- ❑ Bore opening: 85 cm
- ❑ Flat table top



## CT simulator

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The major steps in the target localization and field design are:

### ☐ Physical simulation

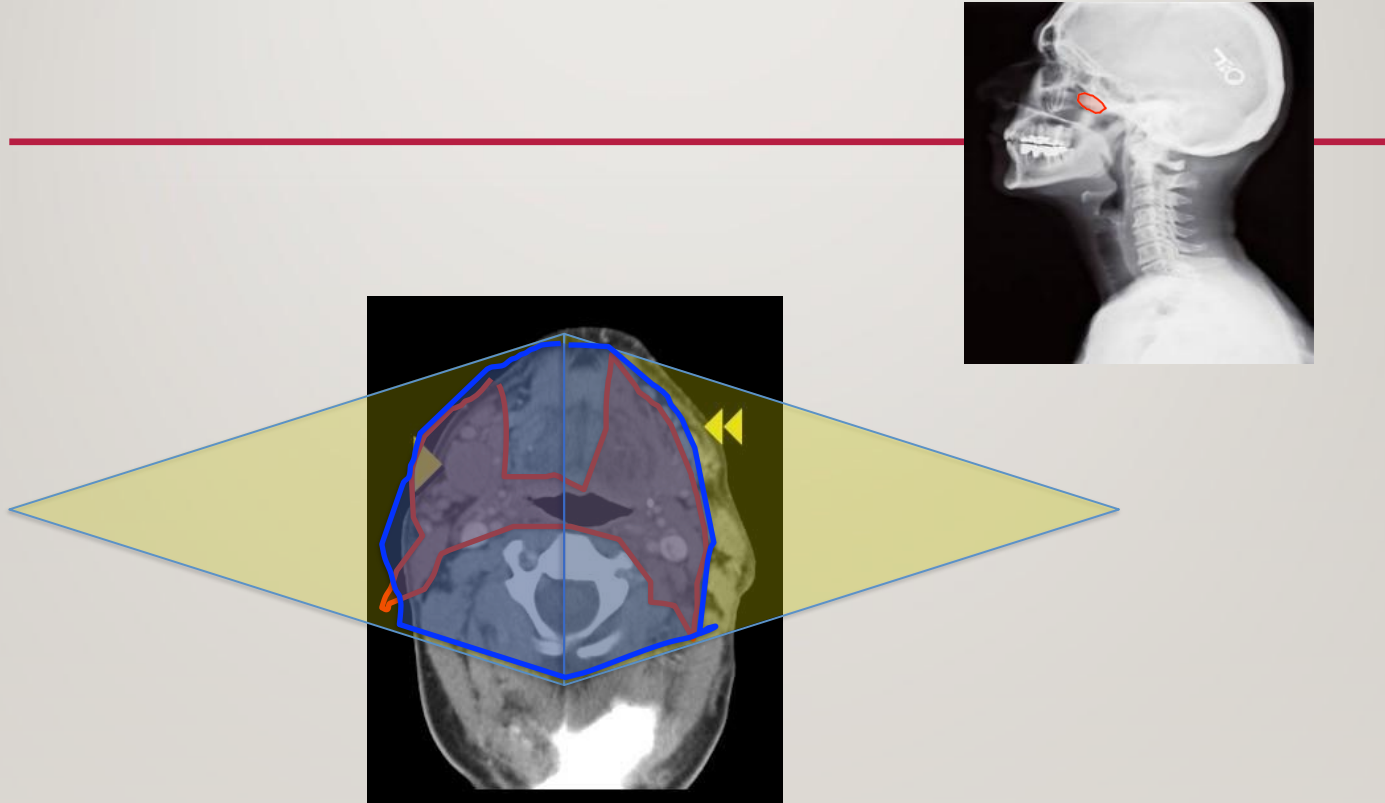
- (1) Acquisition of the patient data set.
- (2) Localization of target and adjacent structures.
- (3) Definition and marking of the patient coordinate system.

### ☐ Virtual simulation

- (1) Design of treatment fields.
- (2) Transfer of data to the treatment planning system (TPS).
- (3) Production of images used for treatment verification.

### 8.3. Different planning for treatment radiotherapy.

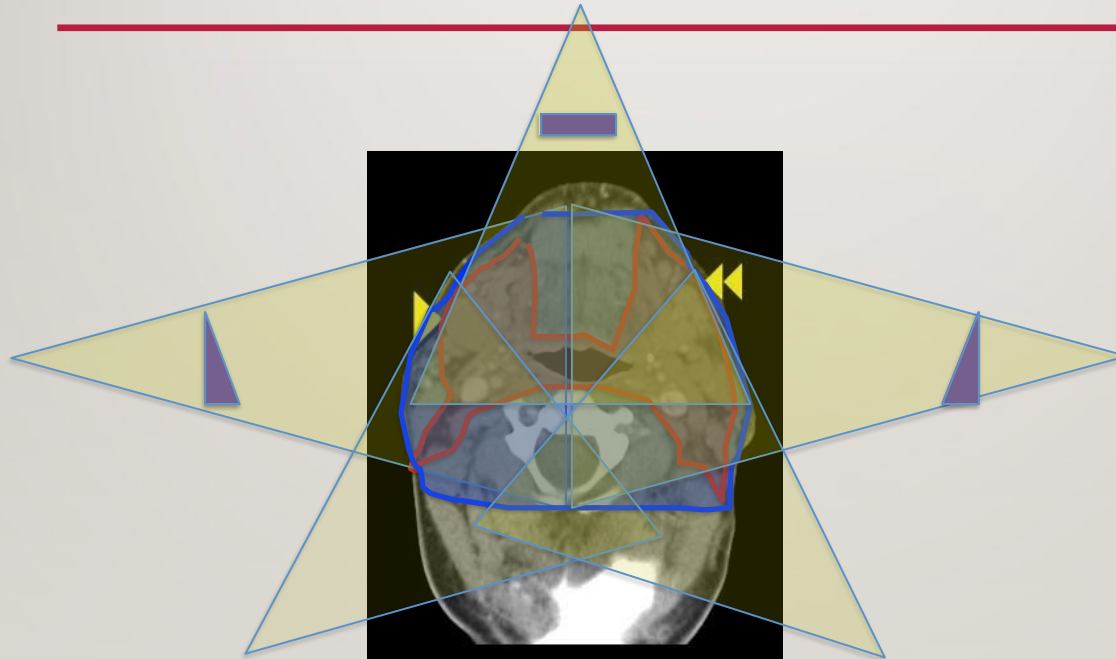
## 2 DIMENSIONAL CONVENTIONAL RADIOTHERAPY





# 3 DIMENSIONAL CONFORMAL RADIOTHERAPY

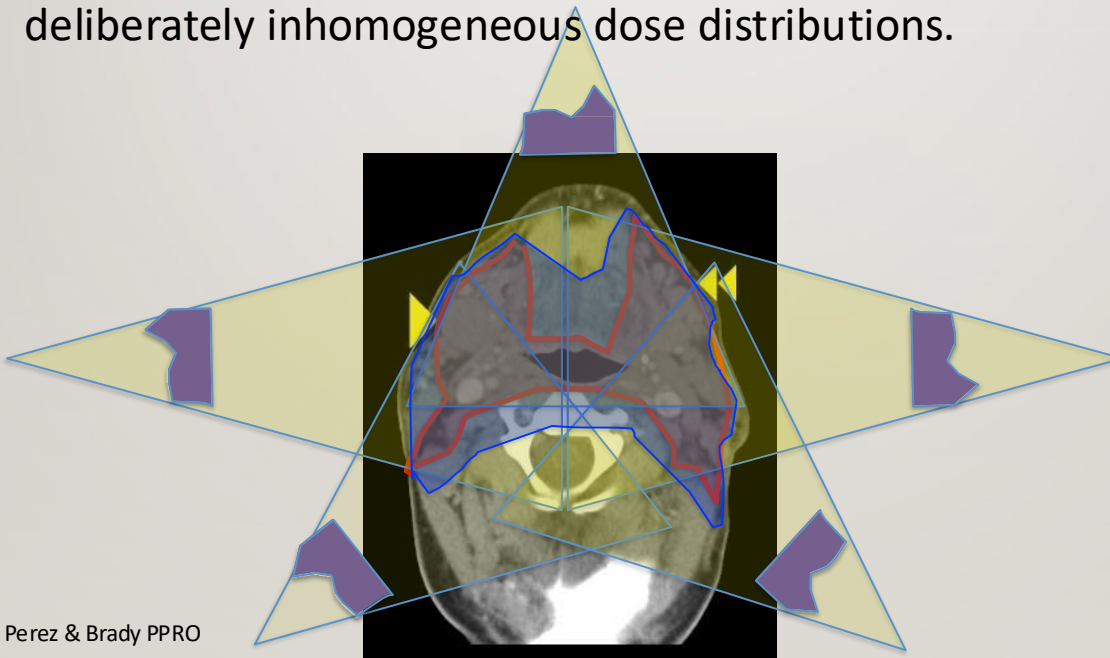
3D conformal radiotherapy (3-D CRT) is the term used to describe the design and delivery of radiotherapy treatment plans based on 3D image data with treatment fields individually shaped to treat only the target tissue.



TECDOC IAEA

# INTENSITY MODULATED RADIOTHERAPY (IMRT)

IMRT is a form of three-dimensional (3D) conformal radiation therapy (CRT) in which a computer-aided optimization process is used to determine customized nonuniform fluence distributions to attain conformation to concave targets and deliberately inhomogeneous dose distributions.



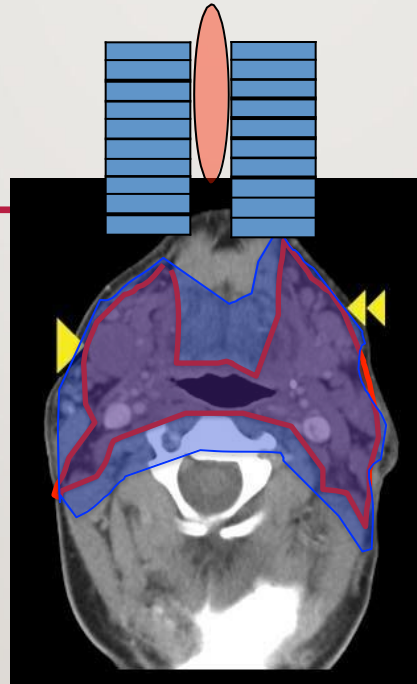
1. Perez & Brady PPRO

# VOLUMETRIC MODULATED ARC THERAPY (VMAT)



“Tomotherapy”

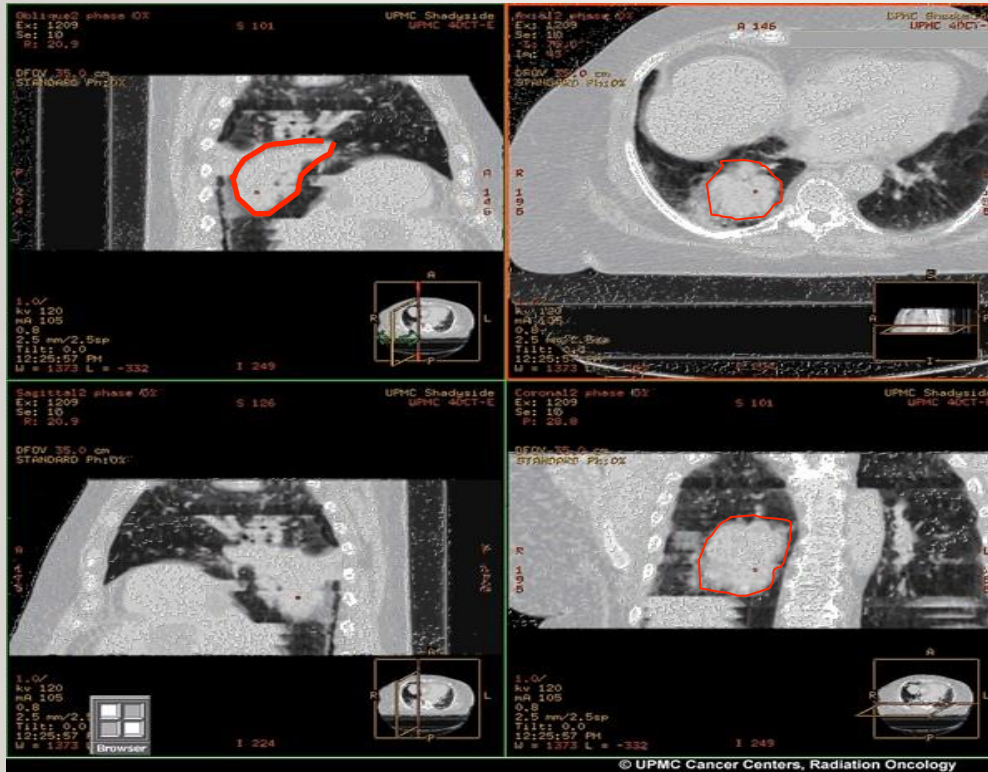
A form of IMRT in which the linear accelerator gantry rotates around the patient while the beam intensity is modulated.



“Rapid Arc”

# Image-guided radiotherapy (IGRT)

## ORGAN MOTION



is the use of modern imaging modalities, to augment target delineation, and the use of imaging to adjust for target motion or positional uncertainty, and, potentially, to adapt treatment to tumor response.

<http://www.upmccancercenters.com>



# STEREOTACTIC RADIOSURGERY



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Stereotactic radiosurgery (SRS) is used for stereotactically 3-dimensional mapping technique) guided conformal irradiation of a defined target volume in a single session

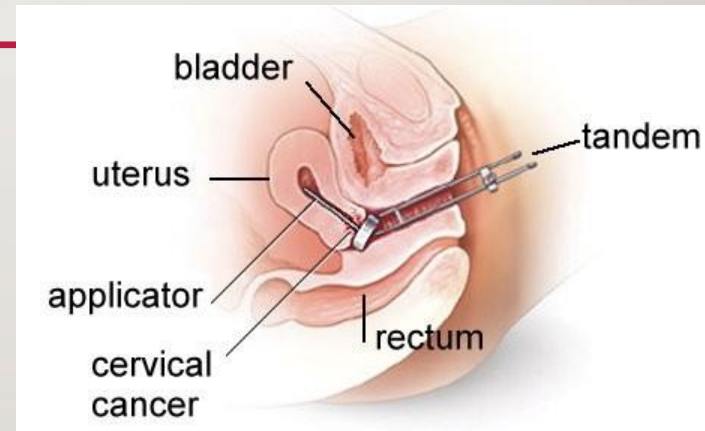
- Gamma Knife Radiosurgery
- LINAC Radiosurgery

1. Perez & Brady PPRO

# BRACHYTHERAPY



Intra-cavitary

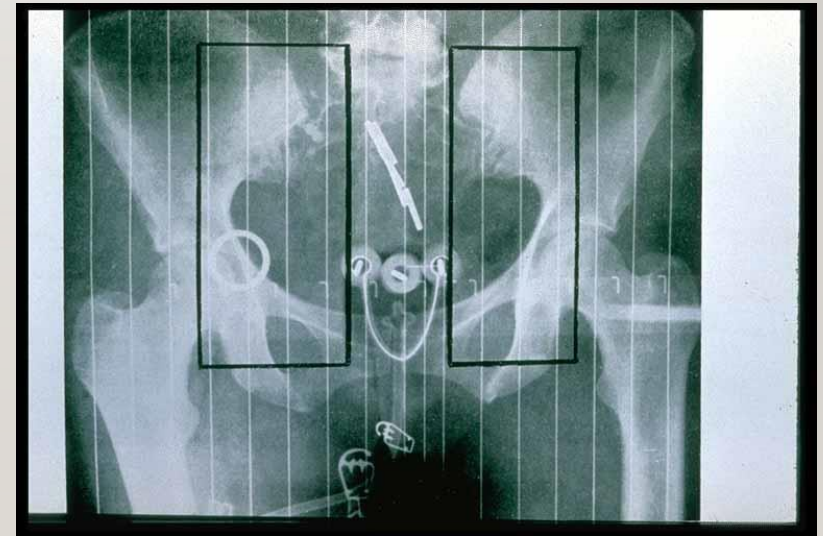


Temporary source

# RADIATION THERAPY

## BRACHYTHERAPY

- This involves placing implants in the form of seeds, wires or pellets directly into the tumour.
- Such implants may be temporary or permanent depending on the implant and the tumour itself.
- The benefit of such a method is that the tumour receives nearly all of the dose whilst healthy tissue hardly receives any.



## QUESTION:

- What are the dose rate ranges for low-, medium-, and high-dose-rate **brachytherapy** according to International Commission for Radiation Units and Measurements (ICRU) Report No. 38?
- 

➤ **Low-Dose-Rate (LDR):**

0.4 to 2.0 Gy per hour—used for permanent and manually after loaders brachytherapy

➤ **Medium-Dose-Rate (MDR):**

2 to 12 Gy per hour. Pulsed dose rate (PDR) brachytherapy after loaders were developed in this dose rate.

➤ **High-Dose-Rate (HDR):**

more than 12 Gy per hour—HDR brachytherapy utilizes very high activity sources.



# RADIOTHERAPY SIDE EFFECTS



## Early (acute reactions)

- Bone marrow
- Oral mucosa
- Intestinal mucosa
- Epidermis (hyperpigmentasi)

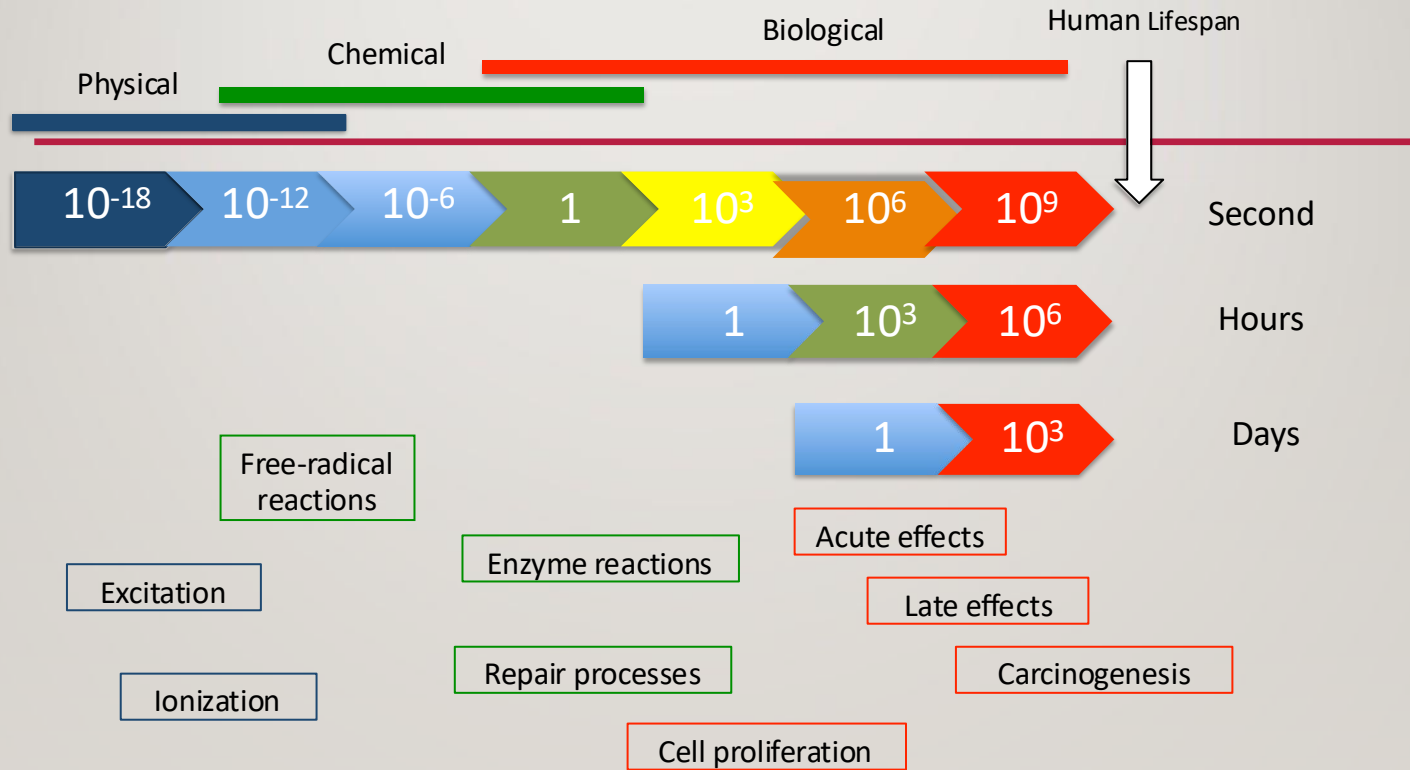


## Chronic (late) responses

- Lung
- CNS
- Kidney
- Dermis
- Parotis (xerostomia)
- Rectal (proctitis)
- Bladder (fibrotic)

90 Days

# WHAT HAPPENS TO THE TISSUE AFTER RADIATION EXPOSURE?



1. Modified from Steel

# RADIO THERAPY

## SIDE EFFECTS



# STERILIZATION

- Radiation not only kills cells, it can also kill germs or bacteria.
- Nowadays, medical instruments (e.g. syringes) are prepacked and then irradiation using an intense gamma ray source.
- This kills any germs or bacteria but does not damage the syringe, nor make it radioactive.



# OCCUPATIONAL EXPOSURE LIMITS (WHOLE BODY)

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|                            |         |
|----------------------------|---------|
| “Trained” Radiation Worker | 5 rem   |
| Pregnant Radiation Worker  | 0.5 rem |
| General Public             | 0.1 rem |

- A rem is a unit of measurement for radiation delivered to human tissue

What level of radiation is safe ??

Consensus of Scientific Opinion:

5 rem annual exposure to an adult

# HOWEVER, THE CONCEPT OF ALARA MUST ALSO BE IMPLEMENTED



## WHAT IS ALARA?

□ AS

□ LOW

□ AS

□ REASONABLE

□ ACHIEVABLE

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## PRINCIPLE OF ALARA

□ MINIMIZE TIME IN RADIATION FIELD

□ MAXIMIZE DISTANCE FROM RADIATION SOURCE

□ USE OF SHIELDING

# MEASUREMENT OF RADIATION EXPOSURE RECEIVED

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- ❑ Personnel Monitoring Device – (External Sources of Radiation).
- ❑ Bioassay –( Internal Radiation)
  - blood sample
  - urine sample
  - thyroid monitoring

# FETAL EXPOSURE CONSIDERATIONS

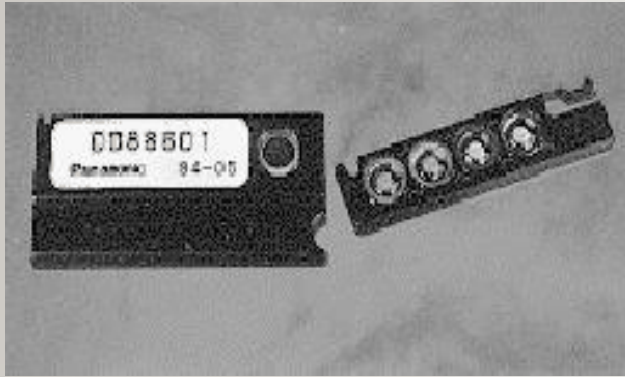
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- First Two Weeks
  - At high doses:  
fetal death
- Weeks 2-15
  - Major organs are developing
  - High doses can cause congenital abnormalities, especially in the brain





# FILM BADGE



# QUESTION:

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- The most recent International Commission on Radiological Protection estimate of the lifetime risk of cancer induction to the general population by low dose rate ionizing radiation is approximately how many cases per Sv?
  - a.  $4 \times 10^4$
  - b.  $4 \times 10^3$
  - c.  $4 \times 10^2$
  - d.  $4 \times 10^1$
  - e.  $4 \times 10^0$

Ans. C

C See ICRU Report #60.

# ANY QUESTIONS?

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